

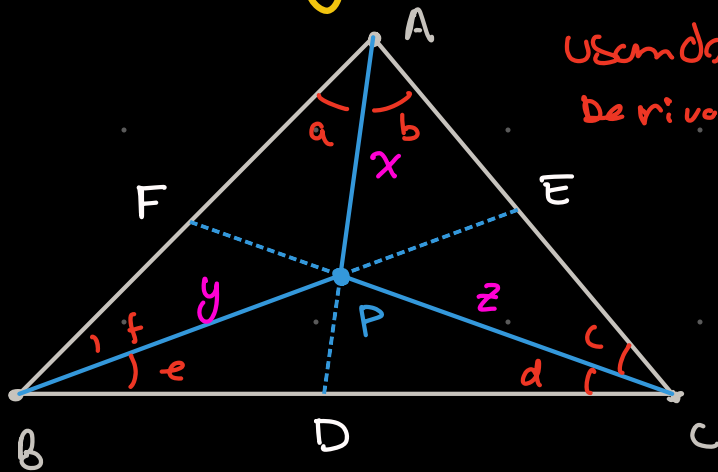
Geometria

Lei dos Cossenos Lei dos Senos

$$a^2 = b^2 + c^2 - 2bc \cos \theta$$

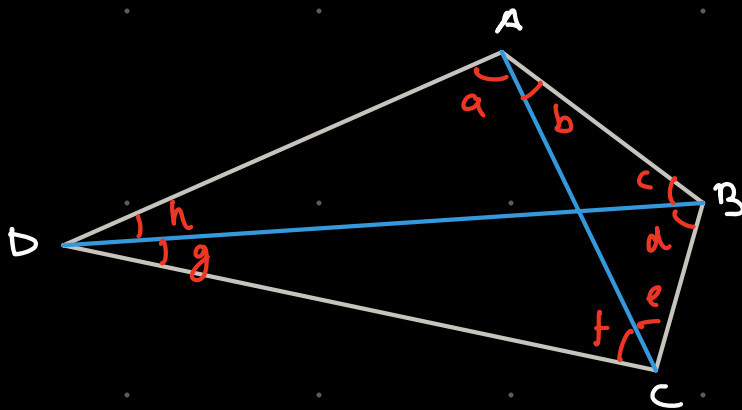
$$\frac{a}{\sin \alpha} = \frac{b}{\sin \beta} = \frac{c}{\sin \gamma} = 2R$$

Ceva Trigonométrico



Usando a Lei dos Senos
Derivando

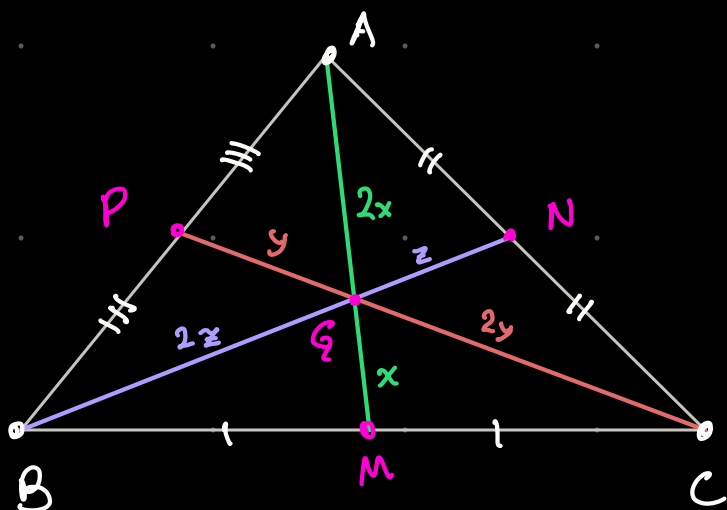
$$\frac{\sin \alpha}{\sin f} \cdot \frac{\sin \gamma}{\sin b} \cdot \frac{\sin e}{\sin d} = 1$$



$$\frac{\sin \alpha}{\sin h} \cdot \frac{\sin g}{\sin f} \cdot \frac{\sin e}{\sin d} \cdot \frac{\sin c}{\sin b} = 1$$

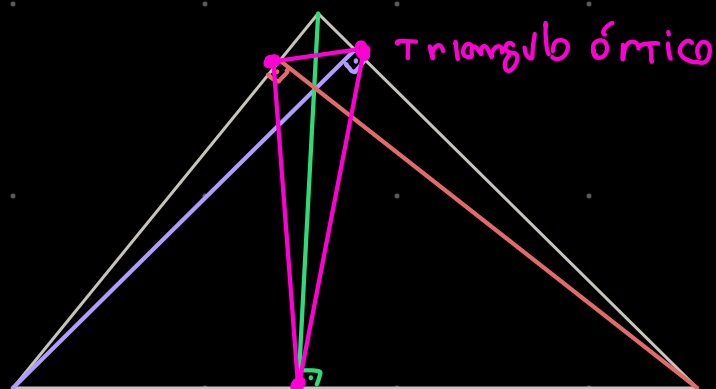
Centros do Triângulo

Baricentro (Medianas)



$$\frac{AP}{PB} \cdot \frac{BN}{NC} \cdot \frac{CM}{MA} = 1 \iff \text{encontro das medianas}$$

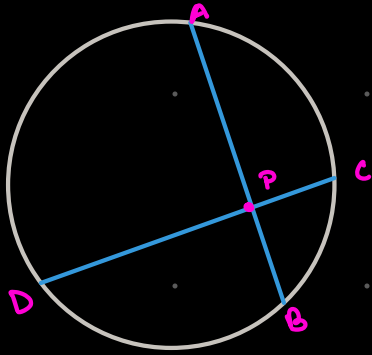
OrtoCentro (Alturas)



encontro das alturas
é centro do órtico

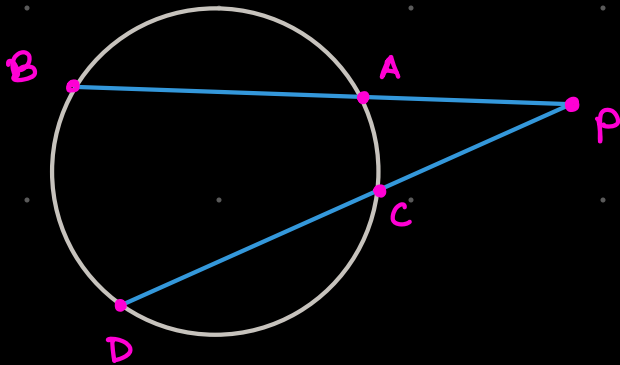
Circunferência

1. Teorema das Cordas



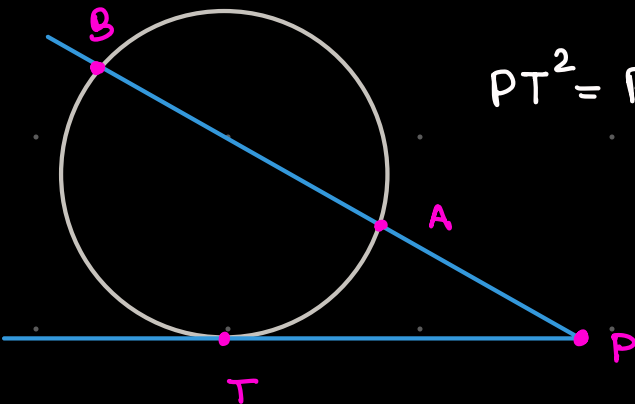
$$PA \cdot PB = PC \cdot PD$$

2. Teorema das Secantes



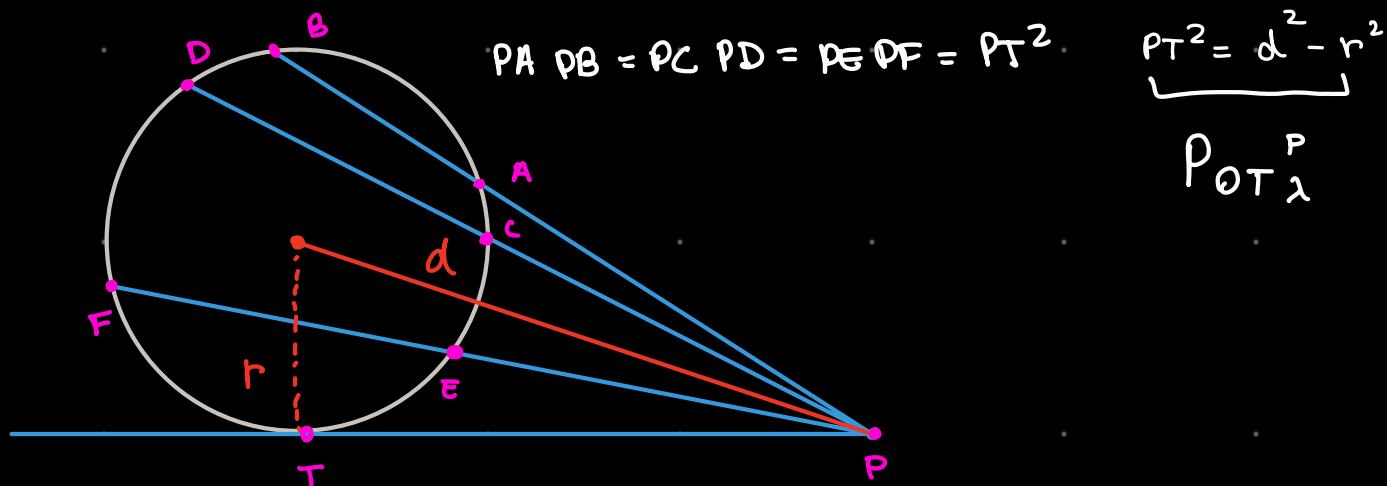
$$PA \cdot PB = PC \cdot PD$$

3. Teorema da Tangente

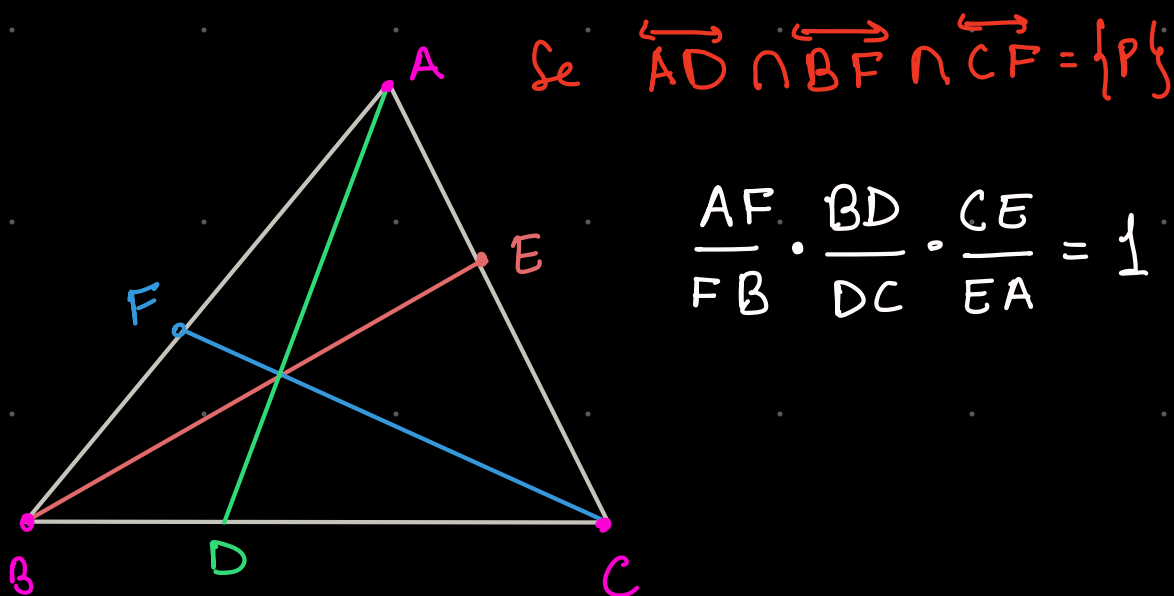


$$PT^2 = PA \cdot PB$$

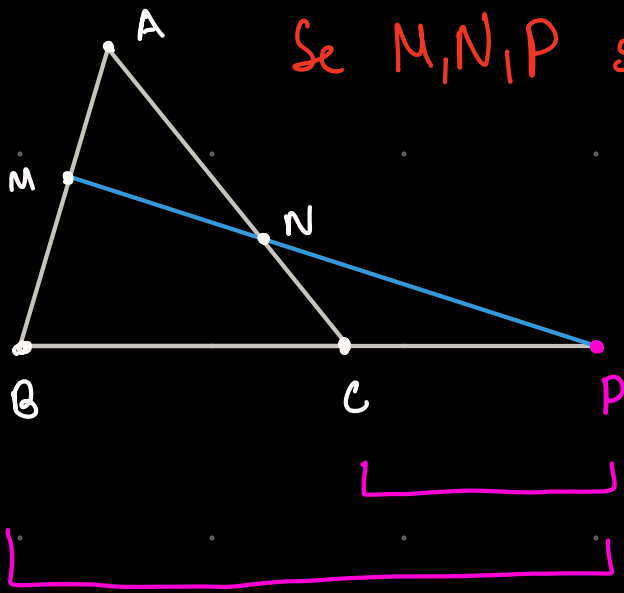
4. Potência de Ponto



Teorema de Ceva



Teorema de Menelaus

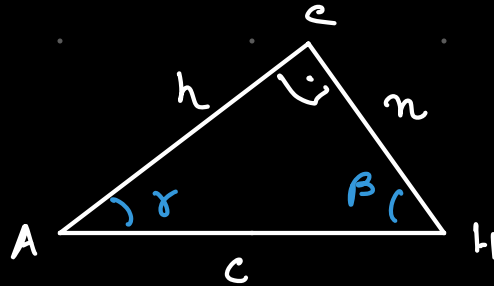
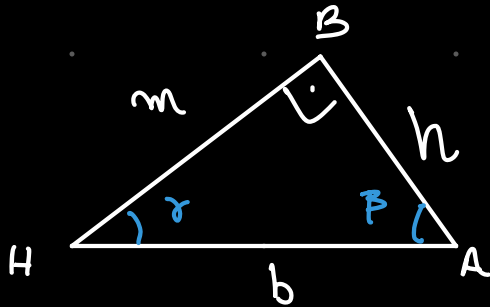
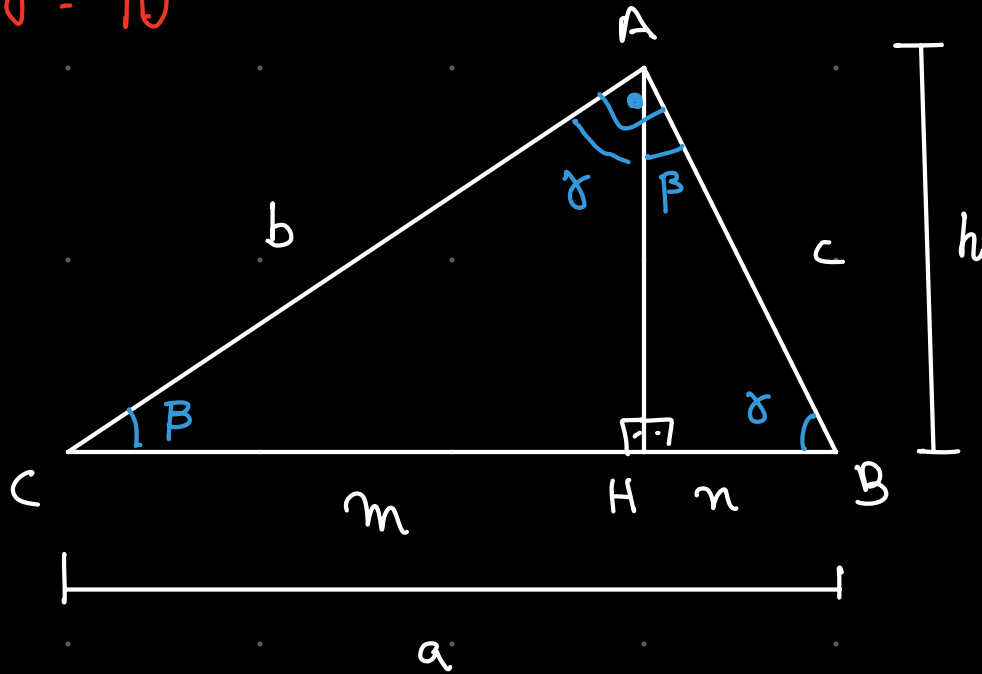


Se M, N, P são Colineares

$$\frac{AM}{MB} \cdot \frac{BP}{PC} \cdot \frac{CN}{NA} = 1$$

Relações Métricas no Triângulo Retângulo

$$\beta + \gamma = 90^\circ$$



$$\triangle ABC \sim \triangle HAC: \frac{a}{b} = \frac{c}{h} \Rightarrow \boxed{a \cdot h = b \cdot c} \quad (i)$$

$$\frac{a}{b} = \frac{b}{m} = \boxed{b^2 = a \cdot m} \quad \begin{array}{l} b \text{ é a média geométrica da} \\ \text{hipotenusa com a sua projeção} \\ \text{sobre ela} \end{array} \quad (ii)$$

$$\triangle ABC \sim \triangle HBA: \frac{a}{c} = \frac{c}{n} = \boxed{c^2 = a \cdot n} \quad (iii)$$

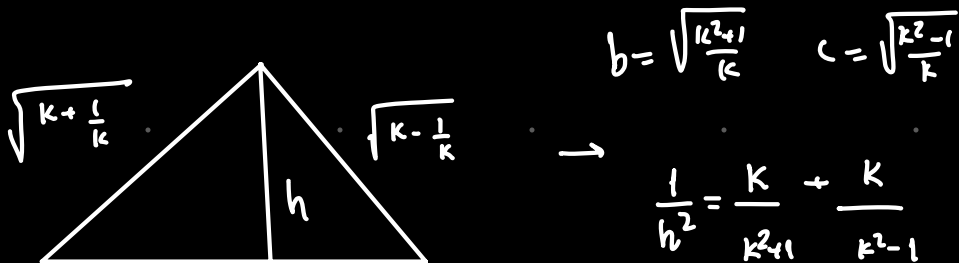
$$\triangle HAC \sim \triangle HBA: \frac{m}{h} = \frac{h}{n} = \boxed{h^2 = m \cdot n} \quad \begin{array}{l} h \text{ é a média geométrica de} \\ m \text{ e } n \end{array} \quad (iv)$$

$$(ii) + (iii) : b^2 + c^2 = a(m+n) \Rightarrow \boxed{a^2 = b^2 + c^2}$$

$$(i) : a \cdot h = b \cdot c \Rightarrow \frac{1}{h} = \frac{a}{bc} \Rightarrow \frac{1}{h^2} = \frac{a^2}{b^2 c^2} \Rightarrow \frac{1}{h^2} = \frac{b^2 + c^2}{b^2 c^2} \Rightarrow \frac{b^2}{b^2 c^2} + \frac{c^2}{b^2 c^2}$$

$$\Rightarrow \boxed{\frac{1}{h^2} = \frac{1}{b^2} + \frac{1}{c^2}}$$

Exercício 13

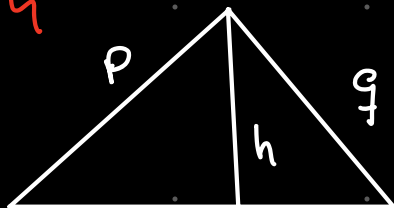


$$b = \sqrt{\frac{k^2+1}{k}} \quad c = \sqrt{\frac{k^2-1}{k}}$$

$$\rightarrow \frac{1}{h^2} = \frac{k}{k^2+1} + \frac{k}{k^2-1}$$

$$\frac{1}{h^2} = \frac{k(k^2-1) + k(k^2+1)}{k^4-1} \Rightarrow \frac{2k^3}{(k^4-1)} \Rightarrow h^2 = \frac{(k^4-1)}{2k^3} \Rightarrow h = \frac{1}{k} \sqrt{\frac{(k^4-1)}{2k}}$$

Exercício 14



$$\frac{2}{p}x^2 - \frac{2}{h}x + \frac{1}{q} = 0 \Rightarrow \Delta = \frac{4}{h^2} - 8 \cdot \frac{1}{q} \cdot \frac{1}{p}$$

c)

$$\frac{1}{p^2} + \frac{1}{q^2} \geq \frac{2}{qp} \rightarrow \frac{q^2 + p^2}{qp} \geq 2$$

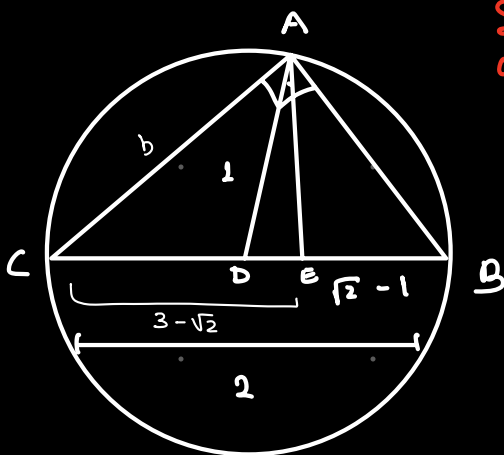
$$q^2 + p^2 \geq 2qp$$

$$q^2 - 2qp + p^2 \geq 0$$

$$(q-p)^2 \geq 0$$

sempre (+)

Ex 16

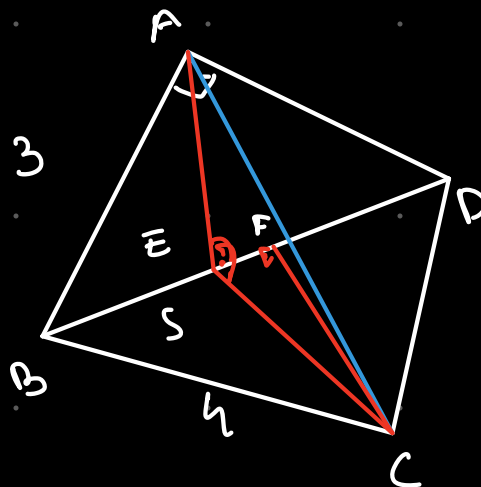
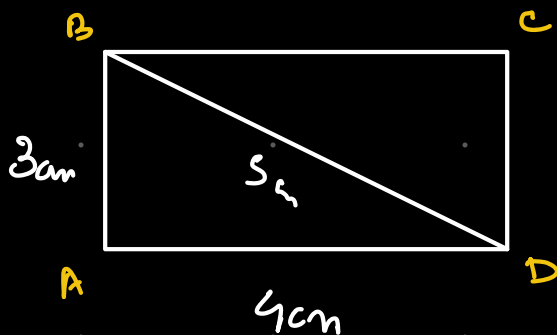


Sempre que ter mediana relativa a hipotenusa de $\triangle$ fazer circunscricao

$$b^2 = 2(3 - \sqrt{2})$$

$$b = \sqrt{6 - 2\sqrt{2}}$$

Ex 15



Falta AC

$$\triangle ADB = \triangle BCD$$

$$AE = CF$$

$$BE = FD$$

$$a \cdot h = bc : 5h = 12 \Rightarrow h = \frac{12}{5}$$

$$AE = CF = \frac{12}{5}$$

$$b^2 = am \rightarrow 5m = 3^2 = m = \frac{9}{5}$$

$$BE = DF = \frac{9}{5}$$

$$c^2 = am \rightarrow 5m = 4^2 = m = \frac{16}{5}$$

$$DE = \frac{16}{5}$$

$$FE = \frac{16}{5} - \frac{9}{5} = \frac{7}{5}$$

$$(ii) \triangle CFE: CE^2 = CF^2 + FE^2 \Rightarrow \left(\frac{12}{5}\right)^2 + \left(\frac{7}{5}\right)^2 = \frac{193}{25}$$

$$CE = \sqrt{\frac{193}{25}}$$

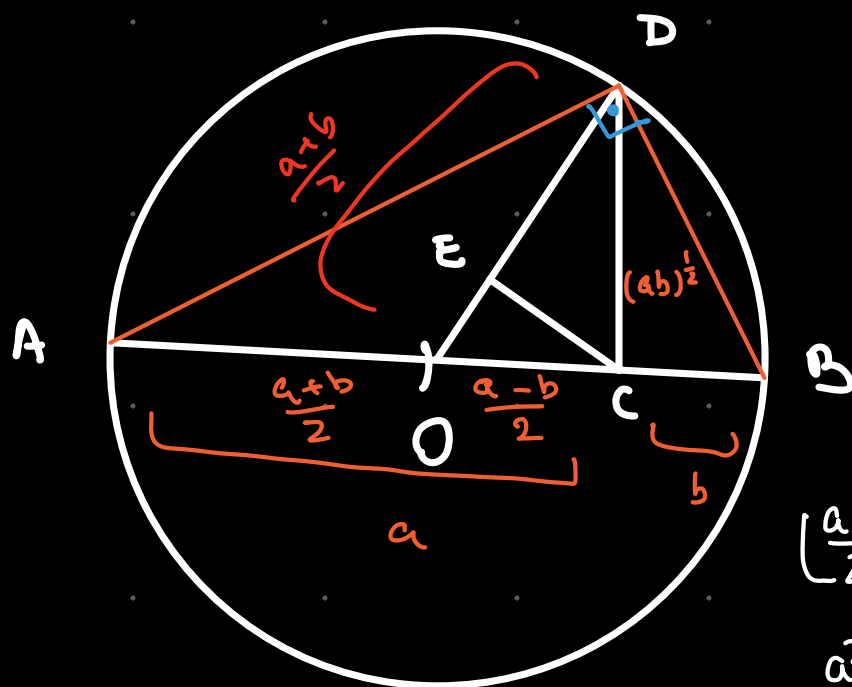
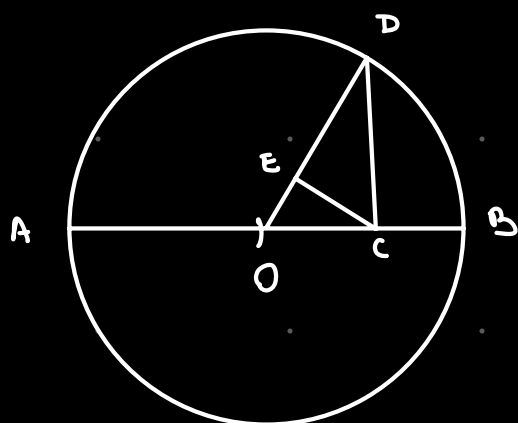
$$(iii) \triangle AEC: AC^2 = AE^2 + CE^2 \quad AC^2 = \left(\frac{12}{5}\right)^2 + \left(\frac{\sqrt{193}}{5}\right)^2$$

$$AC^2 = \frac{144 + 193}{25} \Rightarrow \frac{337}{25}$$

$$AC = \frac{\sqrt{337}}{5}$$

Ex 11. $AC = a$; $BC = b$, O é centro do circ, $CD \perp AB$

$CE \perp OD$



$$\sqrt{a \cdot b}$$

$$\left(\frac{a-b}{2}\right)^2 + ab = \varphi^2$$

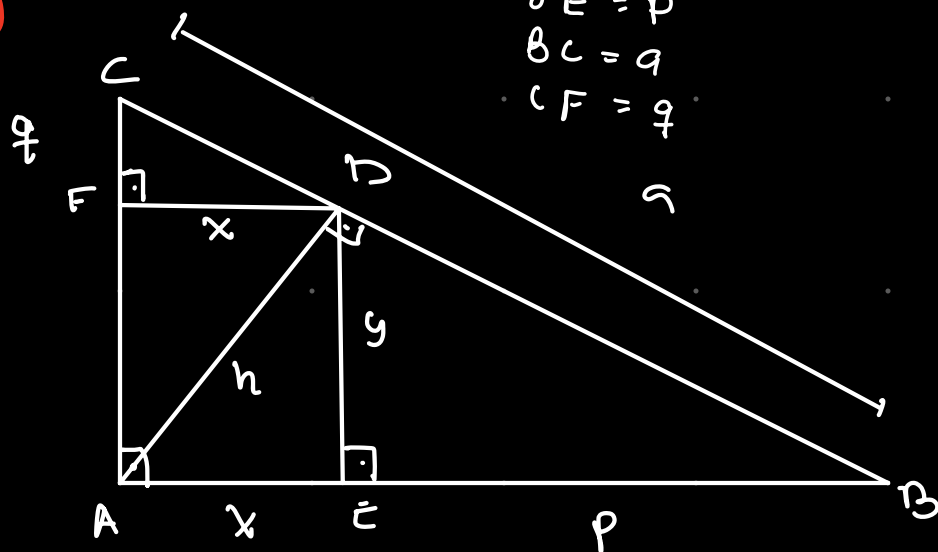
$$a^2 + b^2 - 2ab + 4cb = 4b^2$$

$$\frac{(a+b)^2}{4} \cdot \frac{a-b}{2} = \varphi$$

$$ab = \frac{(a+b)}{2} \cdot n$$

$$\frac{2ab}{a+b} = DE \quad \frac{1}{DE} = \frac{a+b}{2ab}$$

b) φ provada na figura

$$\begin{aligned} AD &= h \\ BE &= p \\ BC &= q \\ CF &= r \end{aligned}$$


$$\Delta CFD : c_D^2 = x^2 + y^2$$

$$\Delta DEB : D_B^2 = y^2 + p^2$$

$$a^2 = CD^2 + DB^2 + 2CD \cdot DB$$

$$\Delta A \in \mathbb{D} : h^2 = x^2 + y^2$$

$$a^2 = x^2 + y^2 + z^2 + 2h^2$$

$$\triangle ABC : CD \cdot BD = h^2$$

$$a^2 = q^2 + p^2 + h^2 + 2h$$

$$a^2 = q^2 + p^2 + 3h^2$$

$$a \cdot h \cdot p_f = n^3 \cdot h$$

$$(AB \cdot AC) p_7 = h^4$$

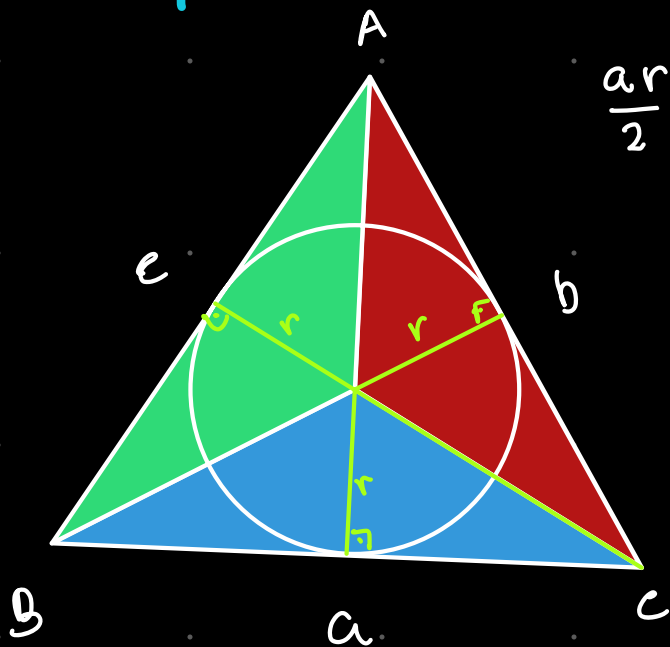
$$(A_B \cdot P)(A_C \cdot Q) = h^4$$

$$PB^2 \cdot CD^2 = h^4$$

$$(DB \cdot CD)^2 = h^4$$

$$DB \cdot CD = h^2 \angle$$

Raio Da Inscrita em Triangulos Gerais



$$\frac{ar}{2} + \frac{br}{2} + \frac{cr}{2} = [ABC]$$

$$r \frac{(a+b+c)}{2} = [ABC]$$

$$rP = [ABC]$$

P é Semiperimetro

$$r = \frac{[ABC]}{P}$$